METAR: Conducting Meta-Analysis with R

Full Program, October 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Conducting Meta-Analysis with R
Format: Online, cohort-based
Duration: 2 weeks, 8 sessions, 2 hours per session, 16 contact hours
Dates: 05/10/2026 – 18/10/2026
Days: Monday, Wednesday, Friday (18h00 to 20h00), Sunday (10h00 to 12h00)
Time: Morocco time
Price: 200€ (2000 MAD)
Instructor: Imad EL BADISY, PhD
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1.1 Course Description

`METAR` provides a practical and methodologically rigorous introduction to conducting meta-analysis with R. It covers the complete evidence-synthesis workflow: formulating a review question, developing an analysis-ready extraction table, computing effect sizes, fitting fixed-effect and random-effects models, assessing heterogeneity, producing forest plots, exploring moderators, assessing publication bias, evaluating certainty of evidence, and preparing a reproducible Quarto report.

The course emphasizes transparent reporting, correct interpretation of pooled estimates, reproducible analysis, and publication-quality tables and figures.

1.2 Target Audience

Clinicians, researchers, PhD students, public health professionals, and evidence synthesis analysts. Basic epidemiology or clinical research experience recommended.

1.3 Prerequisites

Basic familiarity with R (as covered in `BIOSTATR`) is recommended but not mandatory.

1.4 Learning Objectives

By the end of `METAR`, participants will be able to:

1. formulate a meta-analysis question using the PICO framework;
2. structure extracted study data for analysis in R;
3. compute effect sizes for binary and continuous outcomes;
4. fit fixed-effect and random-effects meta-analysis models;
5. quantify and explore heterogeneity using Q, I^2, and tau^2;
6. produce publication-quality forest plots;
7. conduct subgroup analysis and meta-regression;
8. assess small-study effects and publication bias;
9. summarize risk of bias and certainty of evidence;
10. produce a reproducible meta-analysis report with Quarto.

2 Detailed Schedule

Table 1: `METAR` detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	05/10/2026 (Mon)	18h00-20h00	Intro to Systematic Reviews & Meta-Analysis, Protocol and Search Strategy	Evidence hierarchy, PICO, PRISMA workflow, review protocol, eligibility criteria, search strategy, screening
2	1	07/10/2026 (Wed)	18h00-20h00	Data Extraction, Risk of Bias & Preparing Data in R	Extraction tables, risk-of-bias variables, `robvis`, importing/cleaning data, unit harmonization, consistency checks
3	1	09/10/2026 (Fri)	18h00-20h00	Effect Sizes for Binary and Continuous Outcomes	Odds ratio, risk ratio, risk difference, mean difference, standardized mean difference, Hedges' g
4	1	11/10/2026 (Sun)	10h00-12h00	Fixed-Effect and Random-Effects Models	Inverse-variance weighting, common-effect model, DerSimonian-Laird, REML, Paule-Mandel, prediction intervals
5	2	12/10/2026 (Mon)	18h00-20h00	Heterogeneity and Sensitivity Analysis	Cochran's Q, I^2, tau^2, heterogeneity CIs, Baujat plots, leave-one-out analysis, influence diagnostics
6	2	14/10/2026 (Wed)	18h00-20h00	Forest Plots and Result Interpretation	Forest plot anatomy, study weights, summary diamond, prediction interval, subgroup forest plots
7	2	16/10/2026 (Fri)	18h00-20h00	Subgroup Analysis, Meta-Regression, Publication Bias & Certainty of Evidence	Meta-regression, bubble plots, funnel plots, Egger/Begg/Peters tests, trim-and-fill, GRADE overview

No.	Week	Date	Time	Topic	Main Content
8	2	18/10/2026 (Sun)	10h00-12h00	Reproducible Reporting and Final Project	Quarto report, PRISMA diagram, risk-of-bias plot, effect-size table, forest plot

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

Each course ends with a short reproducible Quarto report or mini-project, presented and discussed in Session 8.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.
Software: R ($\geq$ 4.4) and RStudio required.
Attendance: Full attendance at all 8 sessions is recommended given the cumulative structure.